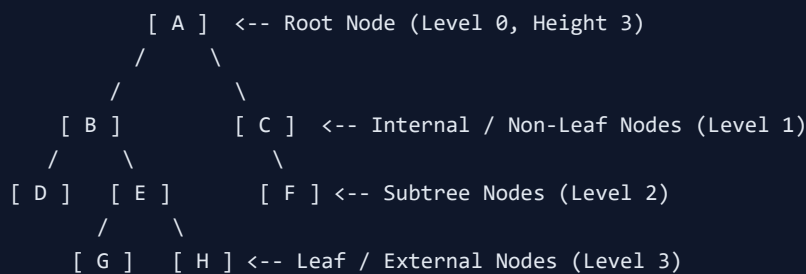


Tree Introduction & Terminologies — Data Structures

Definition: A **Tree** is a non-linear, hierarchical data structure consisting of a collection of nodes connected by directed or undirected edges, such that there exists exactly one path between any two nodes and no cycles are formed.

1. Detailed Technical Explanation

Unlike linear data structures (Arrays, Linked Lists, Stacks, Queues) where elements are stored sequentially, trees organize data hierarchically.



Core Tree Terminologies:

1. **Root:** The topmost node in a tree with no parent (Node **A**).
2. **Edge:** The link or connection between a parent node and its child node.
3. **Parent:** An immediate predecessor node (e.g., **A** is parent of **B** and **C**).
4. **Child:** An immediate successor node (e.g., **B** and **C** are children of **A**).
5. **Siblings:** Nodes that share the same immediate parent (e.g., **D** and **E** are siblings).
6. **Leaf / External Node:** A node with zero children (e.g., **D**, **G**, **H**, **F**).
7. **Internal / Non-Leaf Node:** A node with at least one child (e.g., **A**, **B**, **C**, **E**).
8. **Degree of a Node:** The number of subtrees / children attached to that node.
9. **Degree of a Tree:** The maximum degree of any node in the tree.
10. **Level of a Node:** The distance (number of edges) from the root node. Root is at Level 0 (or Level 1 in some conventions).
11. **Depth of a Node:** The number of edges on the path from the root to that node.
12. **Height of a Node:** The number of edges on the longest downward path from that node to a leaf.
13. **Height of a Tree:** The height of the root node (maximum depth among all nodes).
14. **Subtree:** Any node together with all its descendants forms a subtree.
15. **Path:** A sequence of consecutive edges connecting a sequence of nodes.
16. **Forest:** A set of disjoint trees formed by removing the root node.

2. Memory Keywords & Mathematical Relations

- **Non-linear Hierarchy:** Parent-child recursive relationship.
 - **Node to Edge Relation:** In any valid tree with N nodes, there are exactly $N - 1$ edges.
 - **Path Uniqueness:** There is exactly one unique simple path between any pair of nodes.
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3. Must-Write Points for Exams

- A tree with N nodes always has exactly $N - 1$ edges.
 - Leaf nodes have a degree of 0; internal nodes have degree ≥ 1 .
 - Depth is measured top-down from root (Depth of Root = 0); Height is measured bottom-up from leaf (Height of Leaf = 0).
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4. Quick Recall Flow

Hierarchical Structure -> Root Node -> Edges = $N - 1$ -> Degree (Child Count) -> Height/Depth -> Leaves (Degree 0)